Experiment – 11

Open Ended Project

Aim : To classify legal case documents and generate summaries using NLP, TF-IDF features, and machine learning models.

Theory:

1. **Natural Language Processing (NLP)** is a field of artificial intelligence that enables computers to analyze, understand, and generate human language. It is widely used in applications such as document classification, sentiment analysis, and summarization.
2. **Legal text classification** is an important NLP task where legal documents are automatically categorized into predefined legal articles or topics. In this project, the **ECtHR dataset** is used, which contains legal case descriptions and their associated violated articles.
3. **Text preprocessing** is a crucial step in NLP. It involves cleaning and preparing raw text by applying techniques such as **tokenization, stopword removal, lowercasing, and lemmatization**. These steps reduce noise and improve model efficiency.
4. **Feature extraction** converts textual data into numerical vectors so that machine learning algorithms can process them. The **TF-IDF (Term Frequency–Inverse Document Frequency)** method is used to measure the importance of words in a document relative to the entire corpus. N-grams are also used to capture contextual word relationships.
5. **Multi-label classification** is applied because a legal case may violate multiple legal articles simultaneously. The **One-Vs-Rest strategy** is used to train classifiers for each label.
6. Several machine learning models are used for classification:
    a. **Logistic Regression** – a linear model suitable for text classification.
    b. **Support Vector Machine (SVM)** – effective for high-dimensional text data.
    c. **Naive Bayes** – a probabilistic model commonly used for text analysis.
7. **Text summarization** aims to produce a concise version of a document while retaining key information. Extractive summarization methods like **TextRank, LSA, and Luhn** select the most important sentences from the text.
8. **ROUGE evaluation metrics** are used to measure summarization performance by comparing generated summaries with reference summaries.

Legal Text Classification & Summarization

NLP Minor Project — ECtHR (European Court of Human Rights) Dataset

Project Overview

This notebook covers:

1. **Dataset Loading** — ECtHR NAACL 2021 via KaggleHub
2. **Exploratory Data Analysis (EDA)**
3. **Text Preprocessing** — Tokenization, Stopword Removal, Lemmatization
4. **Feature Extraction** — TF-IDF, BoW
5. **Text Classification** — Logistic Regression, SVM, Naive Bayes
6. **Text Summarization** — TextRank (Extractive)
7. **Evaluation** — Accuracy, F1-Score, ROUGE

Section 1: Install Dependencies

```
!pip install kagglehub nltk scikit-learn sumy rouge-score pandas matplotlib seaborn wordcloud -q
```

```
Preparing metadata (setup.py) ... done
Preparing metadata (setup.py) ... done
Preparing metadata (setup.py) ... done
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 73.5/73.5 kB 2.2 MB/s eta 0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 8.0/8.0 MB 28.7 MB/s eta 0:00:00
Building wheel for rouge-score (setup.py) ... done
Building wheel for breadability (setup.py) ... done
Building wheel for docopt (setup.py) ... done
```

Section 2: Load Dataset

This code downloads the ECtHR NAACL 2021 dataset from Kaggle using the kagglehub library. It then prints the dataset path and lists all files in the directory to verify that the dataset has been downloaded successfully.

```
import kagglehub
import os
import json
import pandas as pd

path = kagglehub.dataset_download("mathurinache/ecthrnaacl2021")
print("Path to dataset files:", path)

for root, dirs, files in os.walk(path):
    for file in files:
        print(os.path.join(root, file))
```

```
Warning: Looks like you're using an outdated `kagglehub` version (installed: 0.3.13), please consider upgrading to the latest version (1.0.0).
Downloading from https://www.kaggle.com/api/v1/datasets/download/mathurinache/ecthrnaacl2021?dataset_version_number=1...
100%|██████████| 32.0M/32.0M [00:00<00:00, 82.5MB/s]Extracting files...

Path to dataset files: /root/.cache/kagglehub/datasets/mathurinache/ecthrnaacl2021/versions/1
/root/.cache/kagglehub/datasets/mathurinache/ecthrnaacl2021/versions/1/train.jsonl
/root/.cache/kagglehub/datasets/mathurinache/ecthrnaacl2021/versions/1/dev.jsonl
/root/.cache/kagglehub/datasets/mathurinache/ecthrnaacl2021/versions/1/test.jsonl
```

This code defines a function **load_jsonl()** to read JSONL files and convert them into a pandas DataFrame for easier data processing. It then identifies the paths for the train, test, and validation datasets within the downloaded ECtHR dataset. Using the **glob** library, the code searches through the dataset directories to locate files containing train, test, and development data and prints their file paths for verification.

```
def load_jsonl(filepath):
    records = []
    with open(filepath, 'r', encoding='utf-8') as f:
        for line in f:
            try:
                records.append(json.loads(line.strip()))
            except:
                pass
    return pd.DataFrame(records)

train_path = os.path.join(path, 'EN_train', 'EN_train')
test_path  = os.path.join(path, 'EN_test',  'EN_test')
dev_path   = os.path.join(path, 'EN_dev',   'EN_dev')

import glob
train_files = glob.glob(path + '/**/*train*', recursive=True)
test_files  = glob.glob(path + '/**/*test*',  recursive=True)
dev_files   = glob.glob(path + '/**/*dev*',   recursive=True)

print('Train files:', train_files)
print('Test files:', test_files)
print('Dev files:', dev_files)
```

```
Train files: ['/root/.cache/kagglehub/datasets/mathurinache/ecthrnaacl2021/versions/1/train.jsonl']
Test files: ['/root/.cache/kagglehub/datasets/mathurinache/ecthrnaacl2021/versions/1/test.jsonl']
Dev files: ['/root/.cache/kagglehub/datasets/mathurinache/ecthrnaacl2021/versions/1/dev.jsonl']
```

This code loads the training, testing, and validation datasets using the **load_jsonl()** function and stores them in separate DataFrames. It then prints the shape of each dataset to show the number of rows and columns. Finally, it displays the first few rows of the training dataset to preview the data.

```
df_train = load_jsonl(train_files[0])
df_test  = load_jsonl(test_files[0])
df_dev   = load_jsonl(dev_files[0])

print(f"Train: {df_train.shape} | Test: {df_test.shape} | Dev: {df_dev.shape}")
df_train.head(3)
```

```
Train: (9000, 12) | Test: (1000, 12) | Dev: (1000, 12)
```

	case_id	case_no	title	judgment_date	facts	applicants	defendants	allegedly_violated_articles	violated_articles	court_assessment_references	silver_ratio
0	001-59587	25702/94	CASE OF K. AND T. v. FINLAND	2001-07-12	[11. At the beginning of the events relevant ...	[K., T.]	[FINLAND]	[13, 8]	[8]	{'8': ['12', '140', '152', '156', '157', '158'...	[1, 13, 21, 2 24, 25, 26, 2 29
1	001-59591	42527/98	CASE OF PRINCE HANS-ADAM II OF LIECHTENSTEIN v.	2001-07-12	[9. The applicant is the monarch of Liechtens...	[PRINCE HANS-ADAM II OF LIECHTENSTEIN]	[GERMANY]	[14, P1-1, 6]	[]	{'6': ['12', '15', '24', '25', '26', '27', '28'...	
2	001-59590	33071/96	CASE OF MALHOUS v. THE CZECH REPUBLIC	2001-07-12	[9. In June 1949 plots of agricultural land o...	[MALHOUS]	[CZECH REPUBLIC]	[6]	[6]	{'6': ['13', '14', '35', '40', '41', '42', '43'...	

Section 3: Exploratory Data Analysis (EDA)

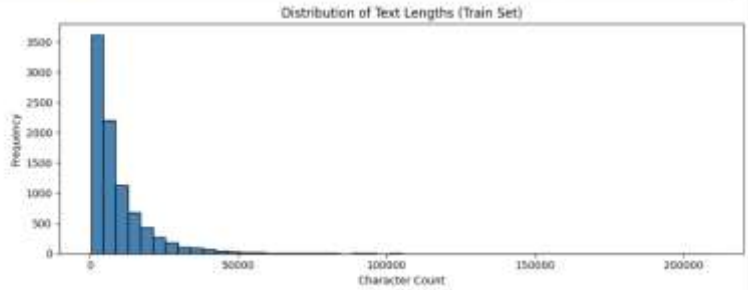
This code imports visualization libraries **Matplotlib** and **Seaborn**, along with **Counter** for analyzing label frequency. It then prints the column names of the training dataset and displays a sample record to understand the structure and contents of the data.

```
import matplotlib.pyplot as plt
import seaborn as sns
from collections import Counter

print("Columns:", df_train.columns.tolist())
print("\nSample record:")
print(df_train.iloc[0])
```

```
Columns: ['case_id', 'case_no', 'title', 'judgment_date', 'facts', 'applicants', 'defendants', 'allegedly_vi

Sample record:
case_id                                                    001-59587
case_no                                                     25702/94
title                                         CASE OF K. AND T. v. FINLAND
judgment_date                                             2001-07-12
facts                        [11.  At the beginning of the events relevant ...
applicants                                                   [K., T.]
defendants                                                  [FINLAND]
allegedly_violated_articles                                  [13, 8]
violated_articles                                                [8]
court_assessment_references    {'0': ['12', '140', '153', '156', '157', '158'...
silver_rationales              [1, 13, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30...
gold_rationales                                                  []
Name: 0, dtype: object
```

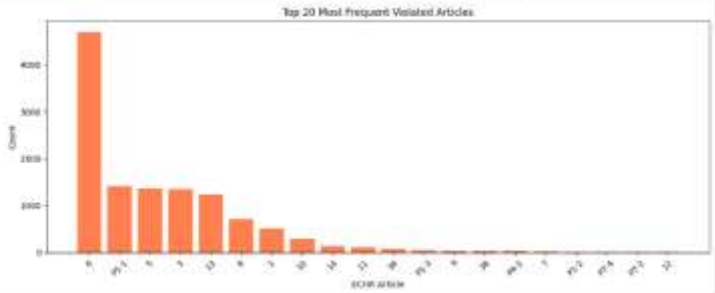
This code combines the paragraphs in the **facts** column into a single text string for each case. It then calculates the length of each text and stores it in a new column called **text_length**. Finally, a histogram is plotted to visualize the distribution of text lengths in the training dataset.

```
df_train['text'] = df_train['facts'].apply(lambda x: ' '.join(x) if isinstance(x, list) else str(x))
df_test['text']  = df_test['facts'].apply(lambda x: ' '.join(x) if isinstance(x, list) else str(x))
df_dev['text']   = df_dev['facts'].apply(lambda x: ' '.join(x) if isinstance(x, list) else str(x))

df_train['text_length'] = df_train['text'].apply(len)
plt.figure(figsize=(10, 4))
plt.hist(df_train['text_length'], bins=50, color='steelblue', edgecolor='black')
plt.title('Distribution of Text Lengths (Train Set)')
plt.xlabel('Character Count')
plt.ylabel('Frequency')
plt.tight_layout()
plt.show()
```



This code analyzes the **violated_articles** column to understand the distribution of legal article labels in the training dataset. It counts how frequently each article appears using the **Counter** function. The top 20 most frequent violated articles are then selected and visualized using a bar chart to show their occurrence in the dataset.

```
label_col = 'violated_articles'

all_labels = [label for sublist in df_train[label_col] for label in sublist]
label_counts = Counter(all_labels)

top_labels = dict(sorted(label_counts.items(), key=lambda x: x[1], reverse=True)[:20])

plt.figure(figsize=(12, 5))
plt.bar(top_labels.keys(), top_labels.values(), color='coral')
plt.title('Top 20 Most Frequent Violated Articles')
plt.xlabel('ECHR Article')
plt.ylabel('Count')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



```
from wordcloud import WordCloud

all_text = ' '.join(df_train['text'].tolist()[:500])  # use subset for speed
wc = WordCloud(width=800, height=400, background_color='white', max_words=200).generate(all_text)

plt.figure(figsize=(14, 6))
plt.imshow(wc, interpolation='bilinear')
plt.axis('off')
plt.title('Word Cloud — Legal Case Facts')
plt.tight_layout()
plt.show()
```

Section 4: Text Preprocessing

This code performs **text preprocessing** on the legal case documents using the NLTK library. It converts text to lowercase, removes special characters, tokenizes the text into words, removes stopwords, and applies lemmatization to reduce words to their base form. The cleaned text is stored in a new column called **clean_text**, which is used for further NLP processing and model training.

```
import nltk
import re
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer
from nltk.tokenize import word_tokenize

nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')
nltk.download('omw-1.4')
nltk.download('punkt_tab')

lemmatizer = WordNetLemmatizer()
stop_words = set(stopwords.words('english'))

def preprocess_text(text):

    text = text.lower()

    text = re.sub(r'[^a-z\s]', '', text)

    tokens = word_tokenize(text)

    tokens = [lemmatizer.lemmatize(t) for t in tokens if t not in stop_words and len(t) > 2]
    return ' '.join(tokens)

print("Preprocessing texts... (this may take a minute)")
df_train['clean_text'] = df_train['text'].apply(preprocess_text)
df_test['clean_text']  = df_test['text'].apply(preprocess_text)
df_dev['clean_text']   = df_dev['text'].apply(preprocess_text)

print("Done! Sample:")
print(df_train['clean_text'].iloc[0][:300])
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
```

Section 5: Label Encoding (for Classification)

This code converts the **multi-label article data** into a numerical format using **MultiLabelBinarizer** from scikit-learn. Each violated article is transformed into a binary vector indicating whether the article is present or not. The transformed labels are stored in **y_train** and **y_test**, and the code prints the label classes along with the shapes of the resulting matrices.

```
from sklearn.preprocessing import MultiLabelBinarizer

mlb = MultiLabelBinarizer()

y_train = mlb.fit_transform(df_train[label_col])
y_test  = mlb.transform(df_test[label_col])

print("Label classes:", mlb.classes_)
print("\ny_train shape:", y_train.shape)
print("y_test shape:",  y_test.shape)
```

```
Label classes: ['10' '11' '12' '13' '14' '18' '2' '3' '34' '38' '4' '46' '5' '6' '7' '8'
 '9' 'P1-1' 'P1-2' 'P1-3' 'P12-1' 'P4-2' 'P4-4' 'P6-1' 'P6-3' 'P7-1'
 'P7-2' 'P7-3' 'P7-4']

y_train shape: (9000, 29)
y_test shape: (1000, 29)
```

Section 6: Feature Extraction — TF-IDF

This code converts the cleaned legal text into numerical feature vectors using the **TF-IDF (Term Frequency–Inverse Document Frequency)** technique. It considers up to 10,000 features and includes both unigrams and bigrams. The vectorizer is fitted on the training data and then used to transform both training and test datasets into TF-IDF matrices for model training.

```
from sklearn.feature_extraction.text import TfidfVectorizer

tfidf = TfidfVectorizer(max_features=10000, ngram_range=(1, 2))

X_train_tfidf = tfidf.fit_transform(df_train['clean_text'])
X_test_tfidf  = tfidf.transform(df_test['clean_text'])

print("TF-IDF Train shape:", X_train_tfidf.shape)
print("TF-IDF Test shape:",  X_test_tfidf.shape)
```

```
TF-IDF Train shape: (9000, 10000)
TF-IDF Test shape: (1000, 10000)
```

This code trains a **Logistic Regression model** for multi-label classification using the **One-vs-Rest strategy**. The model is trained on the TF-IDF feature vectors of the training dataset. After training, it predicts the violated articles for the test data. Finally, the model performance is evaluated using **Micro F1-score and Macro F1-score**, which measure the classification accuracy across all labels.

```
from sklearn.linear_model import LogisticRegression
from sklearn.multiclass import OneVsRestClassifier
from sklearn.metrics import classification_report, f1_score, accuracy_score

print("Training Logistic Regression...")
lr_model = OneVsRestClassifier(LogisticRegression(max_iter=500, C=1.0))
lr_model.fit(X_train_tfidf, y_train)

y_pred_lr = lr_model.predict(X_test_tfidf)

print("\n📊 Logistic Regression Results:")
print(f"  Micro F1 : {f1_score(y_test, y_pred_lr, average='micro'):.4f}")
print(f"  Macro F1 : {f1_score(y_test, y_pred_lr, average='macro'):.4f}")
```

```
Training Logistic Regression...

📊 Logistic Regression Results:
  Micro F1 : 0.6051
  Macro F1 : 0.1517
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: F-score is ill-
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
```

This code trains a **Support Vector Machine (SVM)** model using the **One-vs-Rest classification strategy** for multi-label prediction. The model uses **class_weight='balanced'** to handle class imbalance in the dataset. After training on the TF-IDF features, the model predicts the violated articles for the test data. The performance is evaluated using **Micro F1-score, Macro F1-score, Samples F1-score, and Hamming Loss**.

```
from sklearn.svm import LinearSVC
from sklearn.metrics import hamming_loss

print('Training SVM Balanced...')
svm_bal = OneVsRestClassifier(
    LinearSVC(max_iter=2000, C=1.0, class_weight='balanced')
)
svm_bal.fit(X_train_tfidf, y_train)
y_pred_bal = svm_bal.predict(X_test_tfidf)

print('\n📊 SVM Balanced (class_weight="balanced"):')
print(f'  Micro F1   : {f1_score(y_test, y_pred_bal, average="micro",   zero_division=0):.4f}')
print(f'  Macro F1   : {f1_score(y_test, y_pred_bal, average="macro",   zero_division=0):.4f}')
print(f'  Samples F1 : {f1_score(y_test, y_pred_bal, average="samples", zero_division=0):.4f}')
print(f'  Hamming ↓  : {hamming_loss(y_test, y_pred_bal):.4f}')
```

```
Training SVM Balanced...
```

This code trains a **Naive Bayes classifier** using the **One-vs-Rest strategy** for multi-label classification. The model is trained on the TF-IDF features of the training dataset. After training, it predicts the violated articles for the test data and evaluates the performance using **Micro F1-score and Macro F1-score**.

```python
from sklearn.naive_bayes import MultinomialNB

print("Training Naive Bayes...")
nb_model = OneVsRestClassifier(MultinomialNB())
nb_model.fit(X_train_tfidf, y_train)

y_pred_nb = nb_model.predict(X_test_tfidf)

print("\n📊 Naive Bayes Results:")
print(f"  Micro F1 : {f1_score(y_test, y_pred_nb, average='micro'):.4f}")
print(f"  Macro F1 : {f1_score(y_test, y_pred_nb, average='macro'):.4f}")
```

```
Training Naive Bayes...

📊 Naive Bayes Results:
  Micro F1 : 0.5583
  Macro F1 : 0.1356
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: f-score is ill-
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
```

This code implements **extractive text summarization** using the **Sumy library**. It defines a function that generates summaries using three algorithms: **TextRank, Latent Semantic Analysis (LSA), and Luhn**. The function selects a specified number of important sentences from the original text. A sample legal case from the test dataset is then summarized using all three methods to compare their outputs.

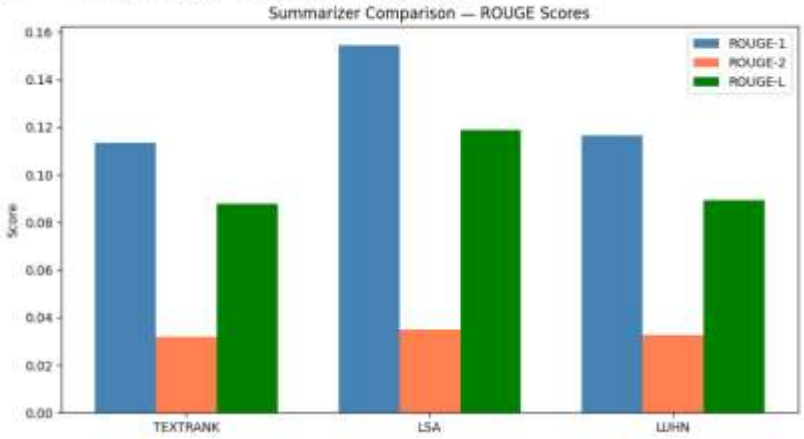
```python
from sumy.parsers.plaintext import PlaintextParser
from sumy.nlp.tokenizers import Tokenizer
from sumy.summarizers.text_rank import TextRankSummarizer
from sumy.summarizers.lsa import LsaSummarizer
from sumy.summarizers.luhn import LuhnSummarizer

def summarize_text(text, summarizer_type='textrank', num_sentences=3):
    """Summarize text using specified algorithm."""
    parser = PlaintextParser.from_string(text, Tokenizer('english'))

    if summarizer_type == 'textrank':
        summarizer = TextRankSummarizer()
    elif summarizer_type == 'lsa':
        summarizer = LsaSummarizer()
    elif summarizer_type == 'luhn':
        summarizer = LuhnSummarizer()

    summary = summarizer(parser.document, num_sentences)
    return ' '.join(str(s) for s in summary)


sample_text = df_test['text'].iloc[0]
print("=" * 60)
print("ORIGINAL TEXT (first 500 chars):")
print(sample_text[:500])
print("\n" + "=" * 60)
print("TEXTRANK SUMMARY:")
print(summarize_text(sample_text, 'textrank', 3))
print("\n" + "=" * 60)
print("LSA SUMMARY:")
print(summarize_text(sample_text, 'lsa', 3))
print("\n" + "=" * 60)
print("LUHN SUMMARY:")
print(summarize_text(sample_text, 'luhn', 3))
```

```
============================================================
ORIGINAL TEXT (first 500 chars):
5.  The applicant is a journalist for DN.no, a Norwegian Internet-based version of the newspaper Dagens Næringsliv ("DN"

============================================================
TEXTRANK SUMMARY:
In the court records ("rettsboken"), his arguments in favour of issuing such an order are restated as having included th

============================================================
LSA SUMMARY:
Other media also reported on the first article, including an online newspaper (Hegnar online) which on 28 August 2007 re

============================================================
LUHN SUMMARY:
The court records contain the following passage: "When heard as a witness [the applicant] stated that she had received A
```

This code evaluates the performance of the summarization algorithms using **ROUGE metrics**. It calculates **ROUGE-1, ROUGE-2, and ROUGE-L scores** by comparing the generated summaries with reference text from the original document. The evaluation is performed on a sample of the test dataset. Finally, the results for **TextRank, LSA, and Luhn** summarizers are displayed and visualized using a bar chart for comparison.

```
from rouge_score import rouge_scorer

scorer = rouge_scorer.RougeScorer(['rouge1', 'rouge2', 'rougeL'], use_stemmer=True)

def evaluate_summarizer(df, summarizer_type='textrank', n_samples=50):
    """Evaluate a summarizer on n_samples from the dataframe."""
    r1, r2, rl = [], [], []
    sample_df = df.head(n_samples)

    for _, row in sample_df.iterrows():
        # Original text as reference (use first 3 sentences as pseudo-reference)
        reference = '. '.join(row['text'].split('.')[:3])
        try:
            hypothesis = summarize_text(row['text'], summarizer_type, 3)
            scores = scorer.score(reference, hypothesis)
            r1.append(scores['rouge1'].fmeasure)
            r2.append(scores['rouge2'].fmeasure)
            rl.append(scores['rougeL'].fmeasure)
        except:
            pass

    return {
        'ROUGE-1': round(sum(r1)/len(r1), 4) if r1 else 0,
        'ROUGE-2': round(sum(r2)/len(r2), 4) if r2 else 0,
        'ROUGE-L': round(sum(rl)/len(rl), 4) if rl else 0
    }

print("Evaluating summarizers (50 samples)...")
results = {}
for method in ['textrank', 'lsa', 'luhn']:
    results[method] = evaluate_summarizer(df_test, method, 50)
    print(f"{method.upper():10s} - {results[method]}")

# Plot ROUGE scores
methods = list(results.keys())
r1_scores = [results[m]['ROUGE-1'] for m in methods]
r2_scores = [results[m]['ROUGE-2'] for m in methods]
rl_scores = [results[m]['ROUGE-L'] for m in methods]
```

```
Evaluating summarizers (50 samples)...
TEXTRANK   - {'ROUGE-1': 0.1134, 'ROUGE-2': 0.032, 'ROUGE-L': 0.0878}
LSA        - {'ROUGE-1': 0.1546, 'ROUGE-2': 0.035, 'ROUGE-L': 0.1188}
LUHN       - {'ROUGE-1': 0.1166, 'ROUGE-2': 0.0328, 'ROUGE-L': 0.0893}
```



This code creates a **summary table of the final results** for both classification and summarization tasks. It stores the performance metrics, including **Micro F1 scores for classification models** and **ROUGE-1 scores for summarization methods**. The results are then organized into a pandas DataFrame and displayed as a table for easy comparison.

```
summary_data_rows = []

summary_data_rows.append({
    'Task': 'Classification',
    'Model/Method': 'Logistic Regression',
    'Micro F1': round(f1_score(y_test, y_pred_lr, average='micro', zero_division=0), 4),
    'Macro F1': round(f1_score(y_test, y_pred_lr, average='macro', zero_division=0), 4),
    'ROUGE-1': None,
    'ROUGE-2': None,
    'ROUGE-L': None
})
summary_data_rows.append({
    'Task': 'Classification',
    'Model/Method': 'SVM',
    'Micro F1': round(f1_score(y_test, y_pred_svm, average='micro', zero_division=0), 4),
    'Macro F1': round(f1_score(y_test, y_pred_svm, average='macro', zero_division=0), 4),
    'ROUGE-1': None,
    'ROUGE-2': None,
    'ROUGE-L': None
})
summary_data_rows.append({
    'Task': 'Classification',
    'Model/Method': 'SVM Balanced',
    'Micro F1': round(f1_score(y_test, y_pred_bal, average='micro', zero_division=0), 4),
    'Macro F1': round(f1_score(y_test, y_pred_bal, average='macro', zero_division=0), 4),
    'ROUGE-1': None,
    'ROUGE-2': None,
    'ROUGE-L': None
})
```

```python
})
summary_data_rows.append({
    'Task': 'Summarization',
    'Model/Method': 'LSA',
    'Micro F1': None,
    'Macro F1': None,
    'ROUGE-1': results['lsa']['ROUGE-1'],
    'ROUGE-2': results['lsa']['ROUGE-2'],
    'ROUGE-L': results['lsa']['ROUGE-L']
})
summary_data_rows.append({
    'Task': 'Summarization',
    'Model/Method': 'Luhn',
    'Micro F1': None,
    'Macro F1': None,
    'ROUGE-1': results['luhn']['ROUGE-1'],
    'ROUGE-2': results['luhn']['ROUGE-2'],
    'ROUGE-L': results['luhn']['ROUGE-L']
})

summary_df = pd.DataFrame(summary_data_rows)
print("\n📊 Final Results Summary")
print(summary_df.to_string(index=False))
```

```
📊 Final Results Summary
         Task        Model/Method  Micro F1  Macro F1  ROUGE-1  ROUGE-2  ROUGE-L
Classification Logistic Regression    0.6051    0.1517      NaN      NaN      NaN
Classification                 SVM    0.6592    0.2463      NaN      NaN      NaN
Classification        SVM Balanced    0.6848    0.3371      NaN      NaN      NaN
Classification         Naive Bayes    0.5583    0.1356      NaN      NaN      NaN
 Summarization            TextRank       NaN       NaN   0.1134   0.0320   0.0878
 Summarization                 LSA       NaN       NaN   0.1546   0.0350   0.1188
 Summarization               Luhn       NaN       NaN   0.1166   0.0328   0.0893
```

Output:

- The ECtHR dataset was successfully loaded and explored using **Exploratory Data Analysis (EDA)**, including label distribution and word frequency visualization.
- After preprocessing and TF-IDF vectorization, the dataset was converted into numerical features suitable for machine learning models.
- Three classification models (**Logistic Regression, SVM, and Naive Bayes**) were trained for **multi-label legal article prediction**, and their performance was evaluated using **Micro and Macro F1 scores**.
- Extractive summarization methods (**TextRank, LSA, and Luhn**) were implemented to generate concise summaries of legal case documents.
- The final pipeline successfully performs **automatic legal case classification and summarization**, demonstrating how NLP techniques can assist in legal document analysis.